

Access Keys & Relationships

Lesson 29 — Keys, Relationships and Integrity

JKSSB Computer Awareness

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Why Keys Matter in a Database

- ▶ A database stores data in **tables**; each table is a grid of rows and columns.
- ▶ A **record** is one row; a **field** is one column holding one kind of value.
- ▶ With many records, the same name or city can repeat, so a row needs a reliable **identifier**.
- ▶ **Keys** provide that identity, and **relationships** join tables that share data.

The one idea

A key identifies a row; a relationship links rows in two tables; integrity keeps those links correct.

The Primary Key

- ▶ A **primary key** (PK) is a field, or set of fields, that **uniquely identifies** each record in a table.
- ▶ No two records may share the same primary-key value — it must be **unique**.
- ▶ A primary key **cannot be left blank** (cannot contain a null value).
- ▶ Each table has **at most one** primary key.

Office example

In an Employee table, EmployeeID can be the primary key because every employee gets a different number.

Composite Key

- ▶ A **composite key** (also called a **compound key**) is a primary key made up of **two or more fields** used together.
- ▶ The combination is unique even if each field alone is not.
- ▶ It is common in a **junction table** that links two other tables.

Office example

In an OrderDetails table, neither OrderID nor ProductID is unique alone, but the pair (OrderID + ProductID) is unique for each line item.

Related key names

A **candidate key** is any field that could be a primary key; an **alternate key** is a candidate key that was not chosen.

The Foreign Key

- ▶ A **foreign key** (FK) is a field in one table that refers to the **primary key** of another table.
- ▶ It is the **link** that creates a relationship between the two tables.
- ▶ Unlike a primary key, a foreign key **may repeat** and may be blank.
- ▶ The table holding the foreign key is called the **child** (related) table.

Trap alert

The foreign key refers to the **primary key** of the other table — not to its own primary key.

Primary Key vs Foreign Key

Point	Primary key	Foreign key
Purpose	Uniquely identifies a record	Links to another table
Uniqueness	Must be unique	May repeat
Blank allowed	No (cannot be null)	Yes
Count per table	At most one	Can be more than one

A table must have a primary key to be the “one” side of a relationship.

Relationships Between Tables

- ▶ A **relationship** works between two tables that share a common field — usually a primary key in one and a foreign key in the other.
- ▶ Access offers three relationship types: **one-to-one**, **one-to-many**, and **many-to-many**.
- ▶ **One-to-many** is the **most common** type in practice.
- ▶ Relationships are drawn and edited in the **Relationships** window.

Why join tables at all?

Splitting data across related tables avoids storing the same value again and again.

One-to-One Relationship

- ▶ In a **one-to-one** relationship, each record in the first table matches **at most one** record in the second table, and vice versa.
- ▶ It is used to split a table with many fields, or to keep sensitive data in a separate table.
- ▶ It is the **least common** of the three types.

Office example

An Employee table and an EmployeeConfidential table, where each employee has exactly one confidential record.

One-to-Many Relationship

- ▶ In a **one-to-many** relationship, one record in the first table matches **many** records in the second table.
- ▶ The “one” side holds the **primary key**; the “many” side holds the **foreign key**.
- ▶ It is the **most common** relationship type used in Access.

Office example

One Department has many Employees, but each Employee belongs to only one Department.

Many-to-Many and the Junction Table

- ▶ In a **many-to-many** relationship, many records in the first table match many records in the second.
- ▶ Access **cannot** store a many-to-many relationship directly.
- ▶ It is split into **two one-to-many** relationships using a third table called a **junction table**.
- ▶ The junction table carries the **foreign keys** of both other tables.

Trap alert

Many-to-many is **not** created directly — it is always implemented through a junction table.

The Junction Table in Detail

- ▶ A **junction table** (also **associative** or **linking** table) connects the two tables of a many-to-many relationship.
- ▶ It contains a **foreign key** for each of the two tables.
- ▶ Its own primary key is usually a **composite key** of those two foreign keys.
- ▶ It can also hold extra fields about the link, such as a quantity or a date.

Office example

Students and Courses are many-to-many; an Enrollments junction table holds StudentID and CourseID (and a grade).

Referential Integrity

- ▶ **Referential integrity** is the rule that a foreign-key value must **match an existing primary-key value** in the related table (or be blank).
- ▶ It stops **orphan records** — a child row pointing to a parent row that does not exist.
- ▶ In Access it is switched on in the **Edit Relationships** / Relationships window.
- ▶ With it switched on, Access supports **Cascade Update** and **Cascade Delete**.

Plain meaning

You cannot add a child record whose parent is missing, and you cannot delete a parent while children still point to it.

Cascade Update and Cascade Delete

- ▶ **Cascade Update**: if the parent primary-key value changes, Access updates the matching foreign-key values automatically.
- ▶ **Cascade Delete**: if a parent record is deleted, Access deletes the matching child records automatically.
- ▶ These options keep the two tables consistent — they are available only when referential integrity is enforced.

Trap alert

Cascade Delete **removes related child records too**. It does not leave them behind.

File Formats: .mdb vs .accdb

Extension	Access version	Notes
.mdb	Access 2003 and earlier	The older format
.accdb	Access 2007 and later	The newer, default format

Trap alert

.mdb is the **old** format; .accdb is the **new** format. Reversing “old” and “new” is the common exam trap.

What .accdb Added

- ▶ **Attachment** fields, **multi-valued** fields, and **calculated** fields.
- ▶ Better integration with SharePoint and Outlook.
- ▶ Improved encryption for the database file.
- ▶ Access 2007 and later can still **open** an old .mdb file, but new databases are saved as .accdb by default.

Trap Alert: Keys, Relationships and Integrity

- ▶ A primary key is **unique** and **not blank**; a foreign key may repeat and may be blank.
- ▶ A table has **at most one** primary key, but can have several foreign keys.
- ▶ One-to-many is the **most common** relationship; many-to-many needs a **junction table**.
- ▶ **Referential integrity** keeps foreign-key values matching primary-key values (TRAP hint).
- ▶ .mdb is **old**, .accdb is **new**.

Lesson 29: Recall

- ▶ **Primary key**: uniquely identifies each record; unique and not blank; at most one per table.
- ▶ **Composite key**: a primary key of two or more fields together.
- ▶ **Foreign key**: a field that refers to another table's primary key; it creates the relationship and may repeat.
- ▶ Relationships: one-to-one, one-to-many (most common), and many-to-many (via a **junction table**).
- ▶ **Referential integrity**: foreign-key values must match existing primary-key values, with Cascade Update / Cascade Delete options.
- ▶ File formats: .mdb is the older Access format; .accdb is the newer default format.